MOVING-WINDOW ENGEL HILBERT COCYCLES WITH $\Omega(\sqrt{k})$ HOMOGENEOUS GAUGE DISTORTION

ARGUS OPEN RESEARCH PROJECT

Abstract. Toward the Hilbert-space Lang–Plaut bi-Lipschitz embedding problem for doubling metric spaces, we construct a family of moving-window injective Hilbert-space cocycles on finite balls of the Engel group that are uniformly doubling, admit an exact logarithmic gauge comparison, and incur distortion at least $c\sqrt{k}$ relative to any fixed homogeneous Engel gauge.

1. Main Construction Theorem

Theorem 1.1. *On the three-step nilpotent Engel group $\mathcal{E}$, there exists a sequence of moving-difference-window maps $F_k : B_{\mathcal{E}}(e, R_k) \to \ell_2$ such that:*

(1) *The induced metric balls are uniformly doubling with doubling constant independent of k.*

(2) *Relative to the homogeneous Carnot–Carathéodory gauge d_{CC}, the compression/distortion of F_k on $B_{\mathcal{E}}(e, R_k)$ is at least $c\sqrt{k}$ for an explicit absolute constant $c > 0$.*

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